



Congressional Stock Trading and Market Outperformance: Evidence from STOCK Act Disclosures

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1 Introduction

Members of Congress have long faced scrutiny over their stock trading activities, with recent polling showing over 80% of Americans support a congressional trading ban (Public Consultation 2024). In 1934 the Securities Exchange Act was passed to limit insider trading, but the rules didn't apply to Congress (Brennan Center for Justice 2024). The Stop Trading on Congressional Knowledge (STOCK) Act of 2012 attempted to remedy this issue, falling short in practice. The Act mandates that Members disclose trades over 1000 dollars within 45 days and explicitly subjects them to insider trading laws, yet the fine for failing to do so is only 200 dollars ("S.2038 – 112th Congress (2011–2012): STOCK Act" 2012). Members of Congress have access to confidential information, legislative power, knowledge of government contracts, and connections to other influential people. The legality of their trading raises questions, especially given examples of suspicious behavior. Shortly before the Affordable Care Act passed, Senator John Kerry, Senator Jim Webb, Congressman Vern Buchanan, as well as many other politicians, purchased healthcare stocks, with estimated gains between 15% to 50% in the following six-month period (Brennan Center for Justice 2024). In another case, the Senate Committee on Health's meeting regarding COVID-19 in January of 2020 led to Senator Dianne Feinstein making significant trades shortly before the market downturn (Brennan Center for Justice 2024). Public trust in Congress remains low, with 7 in 10 Americans holding an unfavorable view (Brennan Center for Justice 2024). While policies have been proposed to ban Members from acquiring new stocks or require full divestment, no legislation has yet passed. Given the opportunity for profit,

examples of possible suspicious trading, and low public sentiment, accusations of wrongdoing are common, but are they justified? Prior research suggests Members of Congress earned significant abnormal returns on their trades before the STOCK Act, but not afterward (Huang and Xuan 2020). Other studies have found that stock ownership correlates with politicians making decisions favorable to those respective companies, suggesting a mutual interest from Members and corporate stakeholders (Tahoun 2014). However, these studies were both published nearly a decade ago and don't account for new data, especially the over \$150 million traded by Congress during the COVID-19 pandemic (Brennan Center for Justice 2024). With 12 years of STOCK Act disclosures now available from 2012 to 2024, this study examines whether U.S. Congressional Members generate abnormal returns on their stock trades, analyzing overall portfolio performance, and whether committee membership generates superior overall returns.

2 Data Cleaning and Preprocessing

For this study, historical time series data was needed for three components. The first component was the Members and their trades, where publicly available data can be collected from disclosures on the official government websites, however the format of the data (PDFs or images of disclosures) made it incredibly difficult to retrieve. Instead an ensemble of three datasets, sourced from Kaggle, provided a comprehensive collection of congressional trades (~70,000 rows). These datasets spanned different ranges between 2012 and 2024, and contained 46,000, 15,000, and 5,000 observations respectively, with many duplicates. However, a limitation with the data sources was the verification of trades with those available on the government websites, as well as differing amounts of information contained, and most notably, naming conventions such as including suffixes or nicknames. The second component was the prices of the stocks themselves, needed for determining profits and losses over holding periods. This data was also sourced from Kaggle as well as the yfinance API. The third component was our benchmark, the S&P 500. This was selected as a baseline due to its historic use as an indication of overall market health, broad scope of sectors, and market capitalization weighting (Reiff 2024). We assume that all the collected data is accurate, and we base our analysis on the closing prices for stocks and the S&P 500.

In order to clean the data, sale and purchase amounts are disclosed as ranges, though some of the datasets only included one bound. These needed to be converted to the disclosure ranges that separate into "1,001–15,000", "15,001–50,000", and so on up to "Over 50,000,000". Since our analysis only focuses on profits, in percentages, the actual purchase amounts could be neglected, but the ranges could be useful for further analysis. To deal with the issue of name mismatches, nicknames were standardized, and all names were reformatted to "First, Last", removing suffixes and occasionally middle names if possible. Over 2,000 tickers failed to download from yfinance and needed to be sourced from the Kaggle dataset or cleaned, with some companies being currently delisted, acquired, or referencing assets from foreign markets. On that point, it is important to note that we are only examining stocks traded on the U.S. market and ETFs, bonds, and foreign assets were not considered. After all the previous cleaning and combining, any persisting duplicates were removed, and purchases and sales of stocks based on senator were paired together (using First In First Out). This left a final dataset of 7,514 observations, consisting of a purchase and sale of a stock, a holding period, and a profit percentage calculated for the duration it stayed in a Members portfolio.

3 Data Analysis

First, returns from the S&P 500 were calculated over the same time period that the stock trades cover, and the compound annual growth rate was calculated to be 12.49%.

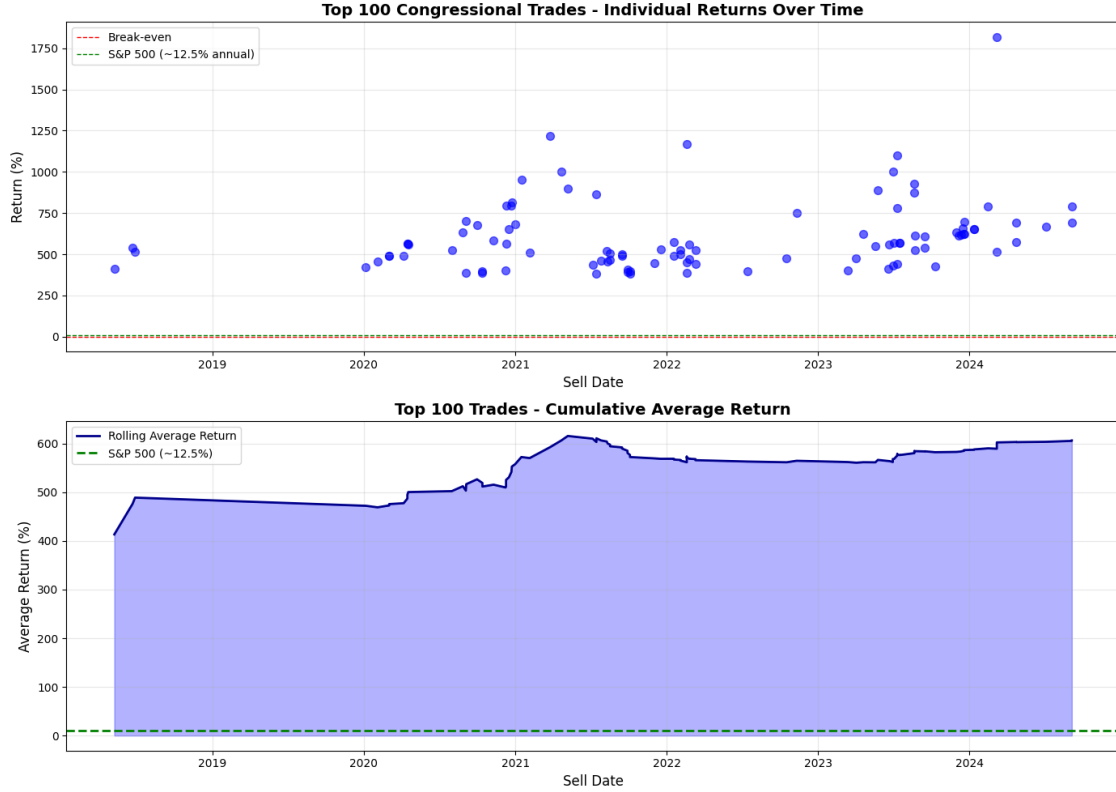


Figure 1: Top 100 Congressional Trades by profit percentage. All trades shown exceed 250% returns. Note that this is just a visualization, and does not have statistical significance.

For each trade, excess returns were calculated by comparing the stock's performance over the holding period to the S&P 500's performance over the same period.

Table 1: Congressional Trades against S&P 500 – Matched Holding Period Analysis

Metric	Value
Number of trades analyzed	7,513
Total trades	7,514
Mean Profit (%)	+32.95
Mean excess return (%)	+13.17
Median excess return (%)	+0.35
Std Dev excess return (%)	81.54
IQR excess return (%)	38.72
25th percentile (%)	-17.81
75th percentile (%)	+20.91
Min excess return (%)	-155.97
Max excess return (%)	+1,702.18
% of trades beating S&P 500	50.7
Wilcoxon test statistic	14,935,624.00
Wilcoxon p-value	0.000006
Significance	***

Table 2: Shapiro-Wilk Test for Normality (Excess Returns)

Metric	Value
Statistic	0.5736
P-value	0.0000
Significance	Not Normal

Table 1 shows the summary statistics for all of the matched congressional trades. Members achieved an average profit percentage of 32.95%, with an average excess of 13.17%. However, we can see large amounts of variability with the median excess at 0.35%, the minimum at -155.97% , and the maximum at 1702.18% . From Table 2, we can see that the excess returns are not normally distributed through a Shapiro-Wilk test with a p-value of 0. Thus a Wilcoxon Signed Rank Test was used to determine significance of the excess returns. At significance level $\alpha = 0.05$, we can see that the excess profits compared to the S&P 500 daily index are highly significant at a p-value of 0.000006.

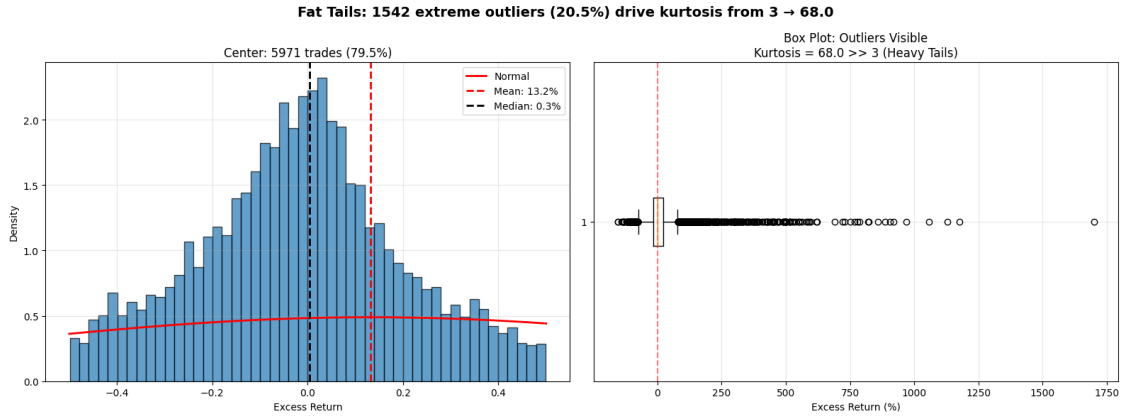


Figure 2: Centered histogram of excess returns (left) and box plot (right) showing skewness and kurtosis of the distribution.

Figure 2 shows the distribution of excess returns has a considerable right skew and extreme kurtosis. The centered histogram (left) shows that the distribution of excess returns is highly non-normal, with a much taller central peak than the normal distribution should have (red line). The box plot (right) confirms this heavy-tailed distribution, with approximately 20.5% of trades being extreme outliers beyond the interquartile range. The incredibly high kurtosis (68) indicates that the average excess returns of 13.7% is not indicative of a typical trade, but is rather pulled by a smaller number of abnormally profitable outliers. This suggests that Members trading results are more extreme than a typical market.

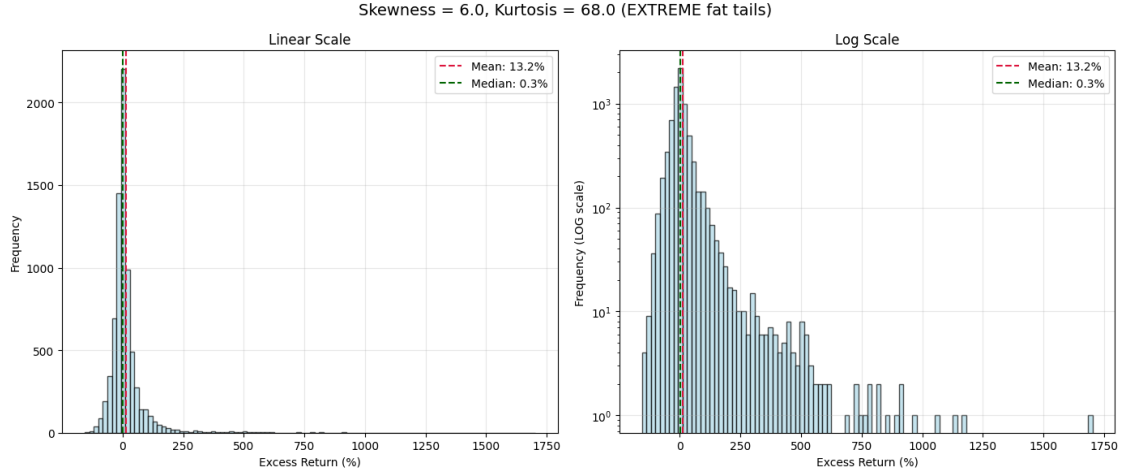


Figure 3: Non-centered excess return distribution on a linear (left) and log (right) scale.

Figure 3 consists of the non-centered excess return distribution on a linear (left) and log (right) scale. The linear scale shows the large concentration around zero, with over 2,000 trades clustered. The log scale makes the long right tail more apparent. This supports the previous discussion of Figure 2.

Table 3: Members with Average Excess Returns Exceeding 100%

Member	Avg Excess Return (%)	Outlier Trades
Warren Davidson	1,177.13	1
James M Inhofe	379.43	10
Shelley M Capito	355.51	46
Robert J Wittman	308.55	1
Gary C Peters	281.50	2
Thomas R Carper	278.92	41
Josh Gottheimer	276.67	100
Pat Roberts	276.07	50
Roger W Marshall	268.60	2
Donald Sternoff Beyer	255.77	5
John Curtis	224.60	12
William Cassidy	218.72	24
Alan S Lowenthal	199.81	10
Jerry Moran	198.31	12
Sheldon Whitehouse	185.85	9
Brian Mast	171.82	2
Patrick J Toomey	171.47	4
Nancy Pelosi	166.16	2
Katherine M Clark	161.69	4
David B Mckinley	159.20	4
Suzan K Delbene	158.31	5
Christopher L Jacobs	152.31	1
Greg Gianforte	152.10	5
Lois Frankel	148.31	10
Gilbert Cisneros	146.81	8
Zoe Lofgren	140.60	2
David A Perdue	135.59	44
Earl Blumenauer	131.13	1
K Michael Conaway	129.23	1
John F Reed	129.03	12
Susie Lee	128.99	6
Mark Green	125.79	25
Thomas H Tuberville	125.07	7
Rick Scott	117.73	1
Ron L Wyden	115.25	7
Garret Graves	115.01	1
Andrew Garbarino	113.69	1
John Rutherford	110.96	3
Dean Phillips	104.28	1
John A Yarmuth	103.25	1
Thomas Suozzi	102.99	1

Table 3 contains Members with average excess returns exceeding 100%. Rep. Warren Davidson stands out for leading the group with an average return of 1,177%, though this is skewed by one outlier trade. Sen. Perdue, Rep. Green, Sen. Carper, Sen. Capito, Sen. Roberts, and Rep. Gottheimer stand out for their large amounts of outlier trades and average excess returns between 125%–350% across 24 to 100 outlier trades.

These large outliers raise concerns that short holding periods may inflate percentage returns, so the dataset was split by holding period. Trades held for at least 365 days were annualized for comparison with the S&P 500’s CAGR, while short-term trades (less than 365 days) were analyzed separately.

Table 4: Annualized Return Statistics (Holding Period ≥ 365 Days)

Metric	Value
Mean	13.46
Median	11.36
Variance	862.59
Std Dev	29.37
Max	651.21
Min	-88.64

Table 5: Short-Term Return Statistics (Holding Period < 365 Days)

Metric	Value
Mean	5.31
Median	1.43
Variance	1,107.99
Std Dev	33.29
Max	511.86
Min	-84.89

Table 4 displays summary statistics for the annualized returns. Members had an average annualized return of 13.45% with a median of 11.36% and a standard deviation of 29.37%. Table 5 shows short term trades had a lower mean and median return of 5.31% and 1.43% respectively, although a higher standard deviation of 33.3%. This suggests that while the longer term holdings consistently perform better, the short term trades are more volatile with most failing to beat the market.

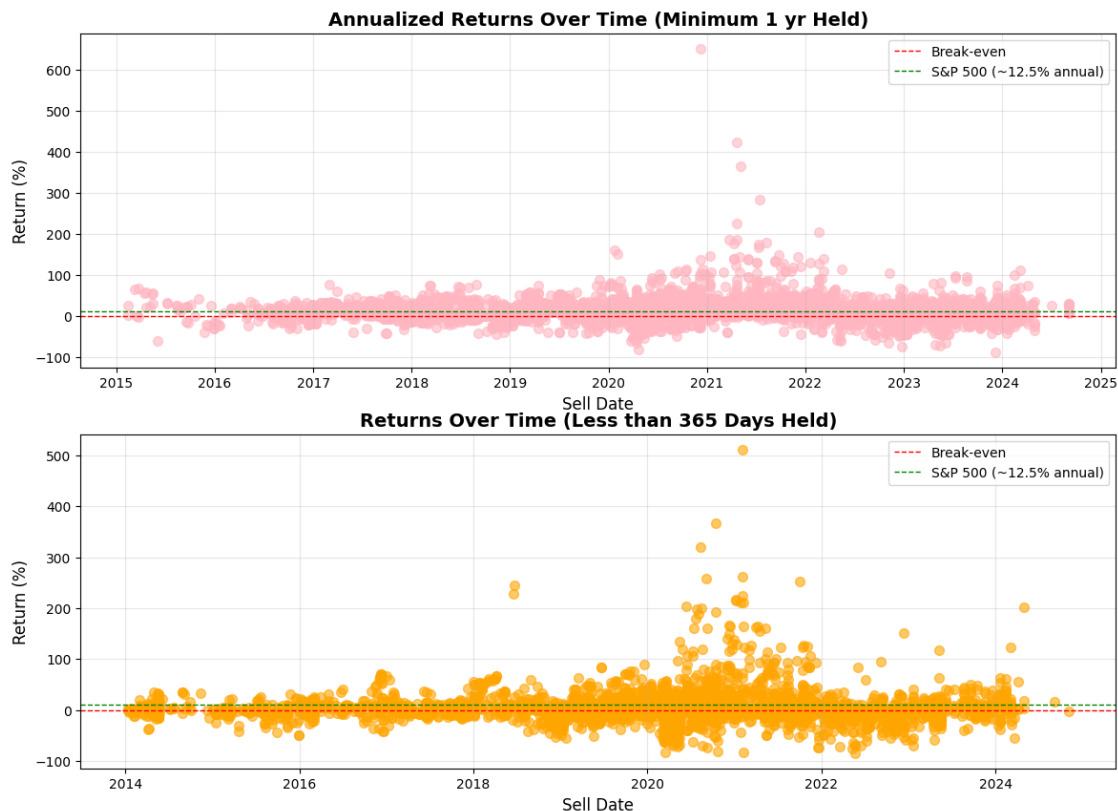


Figure 4: Annualized returns (top) and short-term trades (bottom) plotted over time against the S&P 500 benchmark.

Figure 4 plots the annualized returns and short term trades. Both plots confirm that most trades cluster around the S&P 500 benchmark. Notably, volume and extreme values are most apparent during the pandemic (2020–2022).

3.1 Statistics by Member

From the Members overall portfolios and annualized trades, we get a generalized view of their performance which reveals statistically significant excess returns with a heavily-tailed distribution affected by extreme outliers. To examine if the outperformance is concentrated among a small number of traders, individual member performance can be analyzed.

To focus on Members with sufficient trading activity, this analysis includes only those with at least 5 annualized trades (holding period of at least 365 days), resulting in a subset of 58 Members. Figure 5 visualizes the top 30 performers based on average annualized returns.

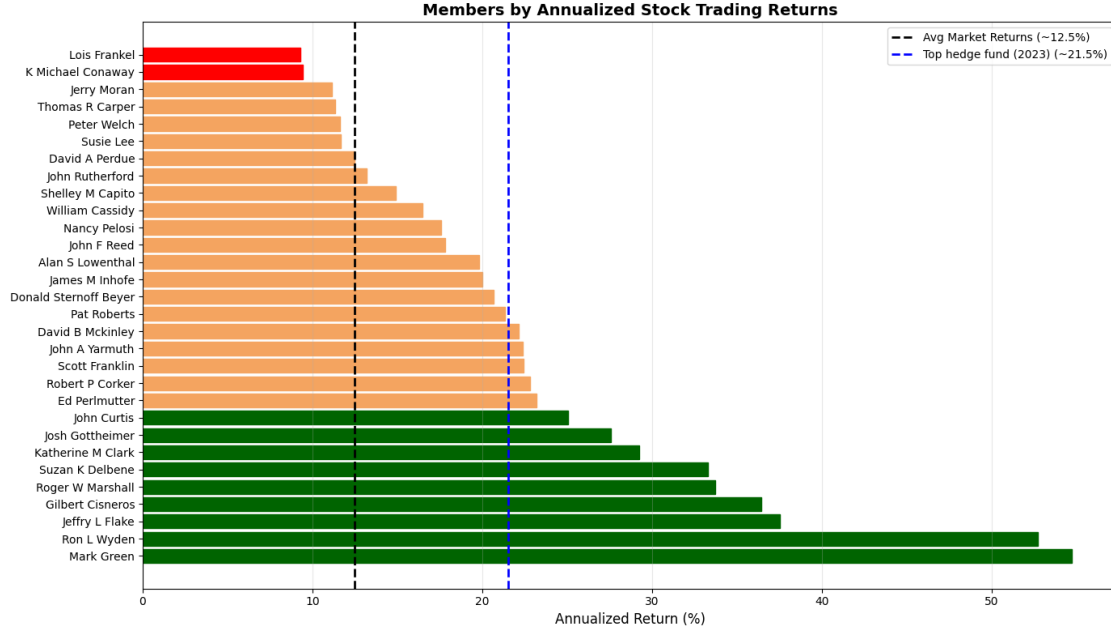


Figure 5: Top 30 Members by average annualized return.

Figure 5 shows that 24 of the top 30 Members or 80% (only 42% of the total 58) beat the S&P 500's CAGR of 12.49% during the same period. Thus, the 6 that did not indicate that the remaining 28 members not included in the top 30 also did not beat the Market benchmark (58% of the total 58). Notably, 14 of the top 15 Members additionally beat the performance of Cotaue Management in 2023 at gains of 21.5%, one of the top performers of the year (Hedge Fund Alpha 2024). While this comparison is not statistically sound, it is useful for interpreting Congressional returns against some of the best professional managers.

A Shapiro-Wilk normality test, as well as a Q-Q plot of the returns was performed on both the trade-level and Member-level annualized returns data. The results are shown below.

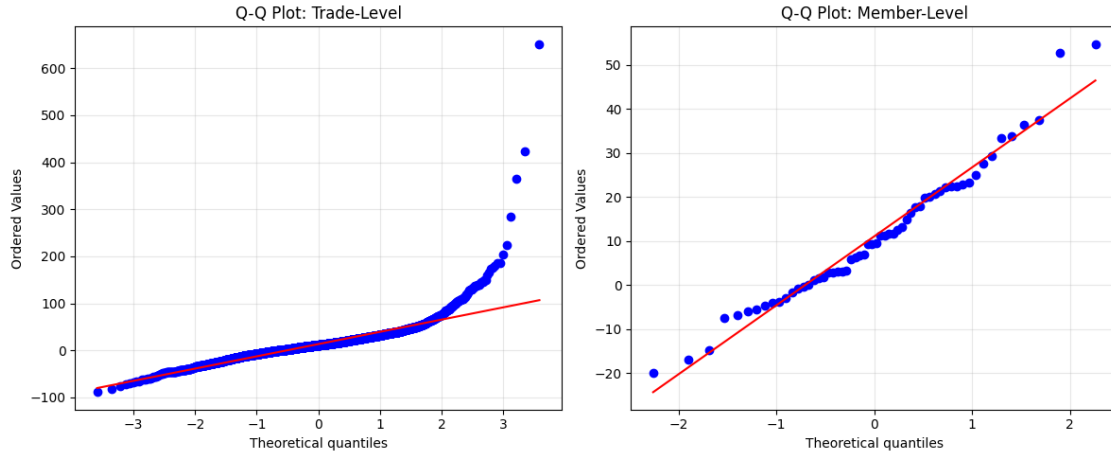


Figure 6: Q-Q Plots for Trade-level (left) and Member-level (right) annualized returns data.

Table 6: Shapiro-Wilk Test for Normality (TL = Trade-level, MEM = Member-level)

Metric	Value
Statistic (TL)	0.7857
P-value (TL)	0.0000
Significance (TL)	Not Normal
Statistic (MEM)	0.9714
P-value (MEM)	0.1866
Significance (MEM)	Normal

From Table 6 and at a significance level of $\alpha = 0.05$, we can observe that the trade-level data is not normally distributed (p-value = 0), while the Member level data is (p-value = 0.18). The Q-Q plots in Figure 6 visualize this, with extreme values in the trade-level (left) exceeding those of a normal distribution, confirming that a small number of exceptional trades influence non-normality. The Member-level plot (right) is close to the theoretical normal line, thus the distribution of Member’s portfolios is approximately normal. This suggests that the individual trades have high variability with large outlier values, but the Members themselves trade with a normal distribution.

Table 7: Trade-level Summary Statistics and Wilcoxon Signed-Rank Test

Metric	Value
Sample size	4,190 trades
Congressional mean return (%)	13.46
S&P 500 benchmark (%)	12.49
Mean excess return (%)	+0.97
Median excess return (%)	−1.13
Excess return volatility (%)	29.37
Wilcoxon test statistic	4,221,965.00
P-value	0.9841
Significance	ns
Effect size (rank-biserial r)	0.0383
Interpretation	Small

Table 8: Member-level Summary Statistics and t -test

Metric	Value
Sample size	58 members
Member mean return (%)	11.05
Member volatility (%)	15.51
S&P 500 benchmark (%)	12.49
Mean excess return (%)	−1.44
Excess return volatility (%)	15.38
t -statistic	−0.7057
P-value	0.7584
Significance	ns
Effect size (Cohen’s d)	−0.0927
Interpretation	Small

Table 9: Comparison of Trade-level and Member-level Annualized Results

Level	N	Mean (%)	Bench. (%)	Excess (%)	Test	P-value	Sig.
Trade	4,190	13.46	12.49	+0.97	Wilcoxon	0.984	ns
Member	58	11.05	12.49	−1.44	t -test	0.758	ns

Table 7 shows that among 4,190 trades held for at least one year, the mean excess return was only 0.97%, which is not statistically significant at $\alpha = 0.05$ (p-value = 0.98). Similarly, Table 8 shows the 58 Members with at least 5 long term trades averaged -1.44% excess returns, was also not significant (p-value = 0.76). These results clarify the findings in Table 1, where we saw a highly significant outperformance of trades against the market. The change in significance when looking at annualized returns shows that the best performance is sourced from short-term trades rather than long-term holdings. This is important to note since if Members were skilled investors, we would expect sustained outperformance across all holding periods. The short-term nature of the successes is more consistent with acting on time-sensitive information or correctly timing market events.

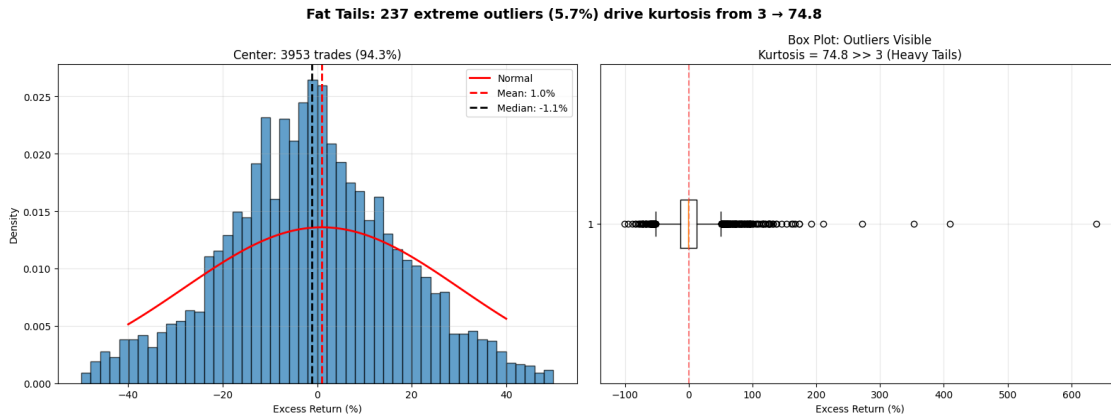


Figure 7: Distribution of annualized excess returns: centered histogram (left) and box plot (right).

Figure 7 shows the distribution of annualized excess returns. The histogram (left) shows that 94.3% of trades fall within 50 percentage points of 0. However, the box plot (right) reveals that outliers making up just 5.7% of the data drive the kurtosis to 74.8. This indicates that while most annualized trades perform near the benchmark, a small subset achieves extremely high or low returns creating the heavy-tailed distribution seen previously.

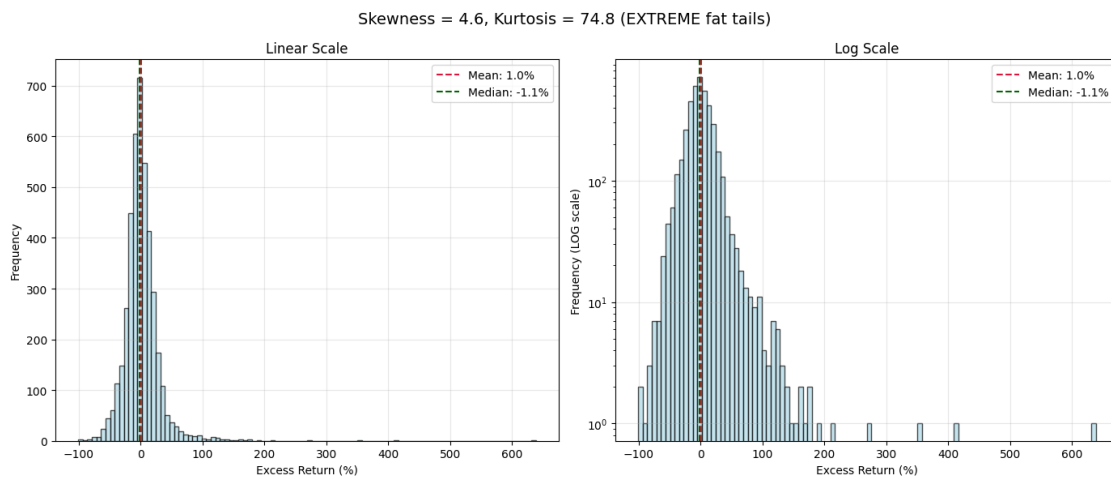


Figure 8: Histogram of annualized excess returns on linear (left) and log (right) scale.

Figure 8 visualizes histogram of the excess returns for the annualized trades, where we can see that most trades are grouped around 0, with the extreme outliers becoming more apparent on the log scale (right).

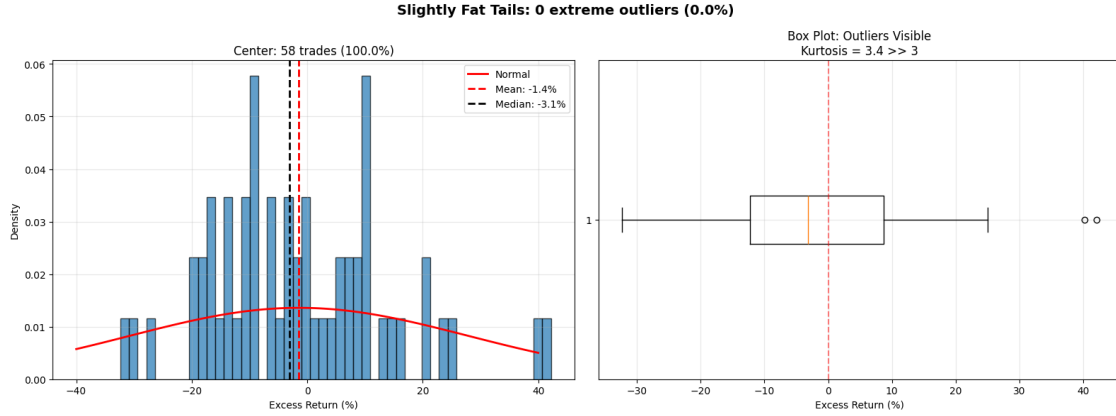


Figure 9: Member-level excess returns: centered histogram (left) and box plot (right).

Figure 9 shows that all 58 Congressional Members fall within 50 percentage points of 0, with no extreme outliers. However, the kurtosis is still slightly larger than what is expected from a normal distribution (3.4 vs 3.0). This is likely due to values that fall within 50 percentage points, but are more extreme like those we can observe around 40 and -30 . This suggests that when annualized trades are grouped by the Members who made them, the excess returns are more normally distributed, which is similar to the findings from Figure 6.

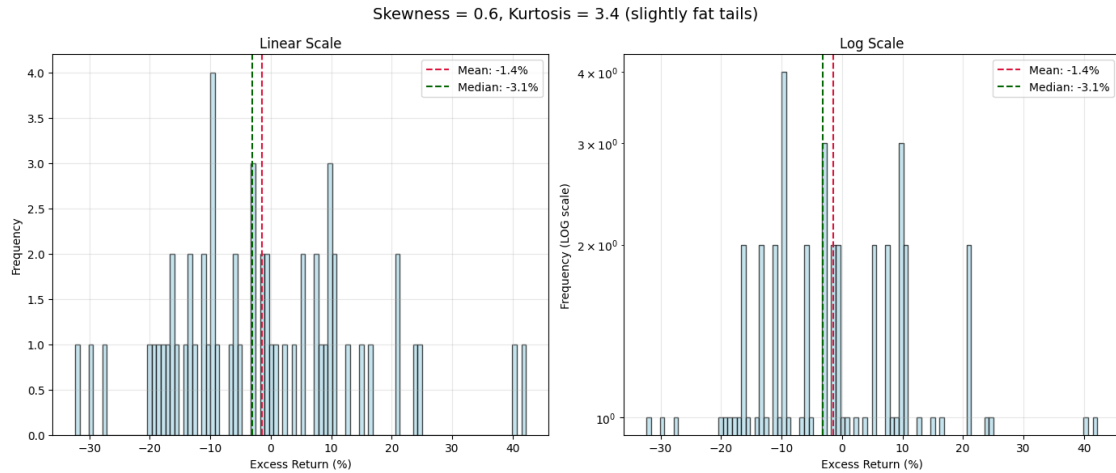


Figure 10: Histograms of Member-level excess returns on a linear (left) and log (right) scale.

Figure 10 shows histograms of the excess returns at the Member-level on a linear (left) and log scale (right). Unlike the trade-level analysis where the log scale drastically changed the visualization, making the heavy-tail more apparent, the log scale here shows minimal difference. We can observe that the data is relatively evenly distributed around the mean. This reinforces previous findings that the Member-level trades have normal excess returns.

It is important to note a difference between the analyses so far. The initial trade analysis in Table 1 compared each trades return to the S&P 500 over that specific holding period. In the annualized returns analysis, the benchmark is a single S&P 500 CAGR of 12.49% calculated over the entire time period from 2012–2024. This means that some periods where Members outperformed during market downturns or underperformed during a bullish period might be less apparent compared to one average CAGR.

3.2 Sharpe Ratios

Table 10: Sharpe Ratio for Monthly Returns from Congressional Trades, 2012–2024 (Held $\geq$ 365 Days)

Metric	Value
Number of months analyzed	147
Months with trades	113
Average monthly return (%)	0.7781
Monthly volatility (%)	1.1627
Risk-free rate (monthly, %)	0.1151
Sharpe Ratio	1.975

Table 11: Sharpe Ratio for S&P 500 Monthly Returns, 2012–2024

Metric	Value
Number of months	145
Average monthly return (%)	1.0622
Monthly volatility (%)	4.1259
Risk-free rate (monthly, %)	0.1151
Annualized return (%)	13.52
Sharpe Ratio	0.795

To assess risk-adjusted performance, Sharpe Ratios for both congressional annualized trades (converted to monthly) and the S&P 500 were calculated, using the 3-month Treasury Bill rate as the risk free rate. Tables 10 and 11 reveal that Members achieved a Sharpe Ratio of 1.975 compared to the S&P 500’s 0.795. This means that congressional trades delivered substantially higher returns for the risk they took on. More specifically, Members earned 1.975 units of excess returns for every unit of risk they take on. Note, that a Sharpe Ratio less than 1 is considered sub-optimal, greater than one is considered good, and over 2 is considered very good (Maverick 2024). This suggests that the Members’ annualized trades are characterized by consistent, lower-risk positions rather than bets on volatile positions. Whether this indicates an information advantage that reduces the uncertainty or conservative portfolio management would require additional research.

3.3 Committees

Previous research in this area discusses Congressional committees and their unique access to a variety of decision-making information. Top “influential” committees were determined from this prior research and from OpenSecrets, who considers influential committees those with the largest revolving door profiles. This refers to members who move between committee positions and private sector roles in the relevant industry. Due to difficulties sourcing historic committee information for all Members in the dataset, at all periods of service, the analysis focuses only on active members and their current committee assignments. There are 39 unique committees with active members in the dataset, and for the purpose of this analysis, only committees with more than 10 combined trades are considered to ensure a more sufficient sample size. Here all Member trades are considered in terms of profit percentage and regardless of holding time (same data as Table 1, not using annualized percentages).

Table 12: Top 20 Committees by Average Profit Percentage

Committee	Avg (%)	Med (%)	Trades	Avg Days
House Perm. Select Cmte. on Intelligence	78.25	28.33	569	729.83
House Cmte. on Financial Services	76.54	24.70	594	732.50
House Cmte. on Foreign Affairs	66.41	-5.87	17	580.53
Senate Cmte. on Homeland Security	58.51	-1.38	15	480.60
Senate Cmte. on Armed Services	55.87	0.23	16	473.06
Senate Cmte. on Commerce, Sci. & Trans.	33.93	16.99	176	828.02
Senate Select Cmte. on Intelligence	33.00	12.79	169	911.32
Senate Cmte. on Indian Affairs	27.04	9.58	201	794.25
Senate Cmte. on Agriculture	25.69	7.84	221	757.31
SC Between US and CCP	24.23	15.92	37	541.46
House Cmte. on Appropriations	23.85	16.44	578	596.30
Senate Cmte. on Veterans' Affairs	20.26	2.70	269	507.66
House Cmte. on Natural Resources	19.08	12.73	383	468.29
Senate Cmte. on Appropriations	16.27	1.15	375	553.54
Senate Cmte. on Finance	14.69	1.83	267	888.12
US Senate Caucus on Int'l Narcotics	14.67	1.68	249	921.10
Senate Cmte. on the Judiciary	14.52	1.68	269	881.80
Comm. on Security & Cooperation in Europe	14.34	0.39	250	914.32
Senate Cmte. on Environment & Public Works	14.34	0.39	250	914.32
House Cmte. on Armed Services	12.08	3.28	81	350.98

From Table 12 we can observe that the best performing committee is the House Permanent Select Committee on Intelligence with a 78.25% average profit and a 28.33% median. Maintaining this profitability with over 500 trades could suggest possibly information advantages. Other committees like the House Committee on Financial Services, House Committee on Foreign Affairs, and Senate Committees on Foreign Affairs, Governmental Affairs and Armed Services all stand out for beating the market benchmark by over 30%. Furthermore, 19 of the top 20 committees beat the market benchmark by at least a percentage point. Out of 39 total committees analyzed this means 48.7%, nearly half of all committees, beat the market benchmark.

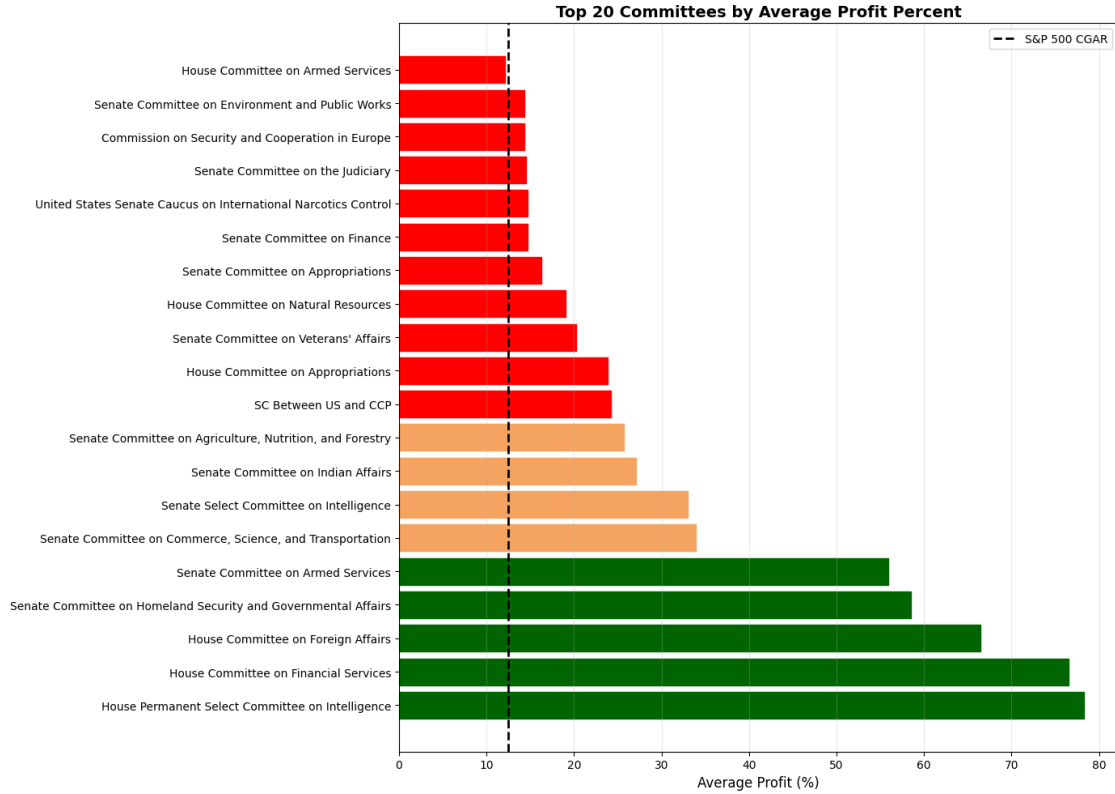


Figure 11: Top 20 committees by average profit percentage.

Figure 11 visualizes Table 12, so no new findings are apparent. However, it is worth noting the large discrepancy in the top 5 committees compared to the rest.

Before determining significance of committee level trades, it is important to understand their distribution.

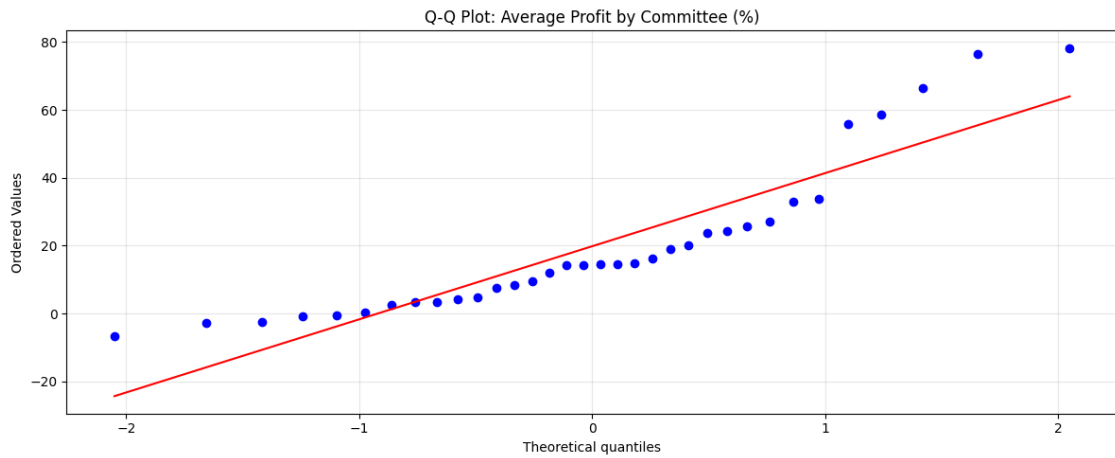


Figure 12: Q-Q Plot for Committee Level Trades.

Table 13: Shapiro-Wilk Results for Committee Level Trades

Metric	Value
Statistic	0.8394
P-value	0.000164
Significance	Not Normal

Figure 12 shows that the average profit by committees appears roughly normal until theoretical quantiles 1–2 where the data deviates from the normal line more extremely. These 5 extreme data points could indicate the 5 highest performing committees that all beat the market benchmark by over 30%. The Shapiro-Wilk test confirms that the trades by committee are not normally distributed at a significance level of $\alpha = 0.05$ and a p-value of 0.000164.

Table 14: Committee-level Wilcoxon Signed-Rank Test Results

Metric	Value
Num Committees	34
Sample size (trades)	2,202
S&P 500 benchmark (%)	12.49
Mean excess return (%)	+7.34
Median excess return (%)	+1.94
Excess return volatility (%)	22.32
Wilcoxon test statistic	352.00
P-value	0.1757
Significance	ns
Effect size (rank-biserial r)	−0.1832
Interpretation	Small

Table 14 shows the results of a Wilcoxon-Signed rank test on the excess returns by committee to be not significant at $\alpha = 0.05$ with a p-value of 0.176. With 34 committees considered, and 2,202 trades between them, the committees only had a mean excess return of 7.34%. These returns are more recent as these are actively serving Members and their current committee assignments (as of November 2025). While Table 12 indicated that high committee returns could indicate significance, we still observed that about half the committees failed to beat the benchmark. Furthermore, Table 14 shows the volatility is 22.32%. This suggests a wide range of success with top performers seeing outstanding gains while others underperform.

Additionally, top influential committees can be analyzed to see if those perceived to have access to more actionable information do produce significant excess returns. These committees were referenced by Ahmed Tahoun in his 2013 paper “The role of stock ownership by US members of Congress on the market for political favors”(Tahoun 2014), as well as from OpenSecrets(OpenSecrets 2024).

Table 15: Top 20 Committees Identified by Tahoun – Summary Statistics and Significance

Metric	Value
Num Committees	20
Sample size (trades)	2,177
S&P 500 benchmark (%)	12.49
Mean excess return (%)	+12.21
Median excess return (%)	−3.12
Excess return volatility (%)	96.64
Wilcoxon test statistic	1,135,336.00
P-value	0.956
Significance	ns
Effect size (rank-biserial r)	0.0422
Interpretation	Small

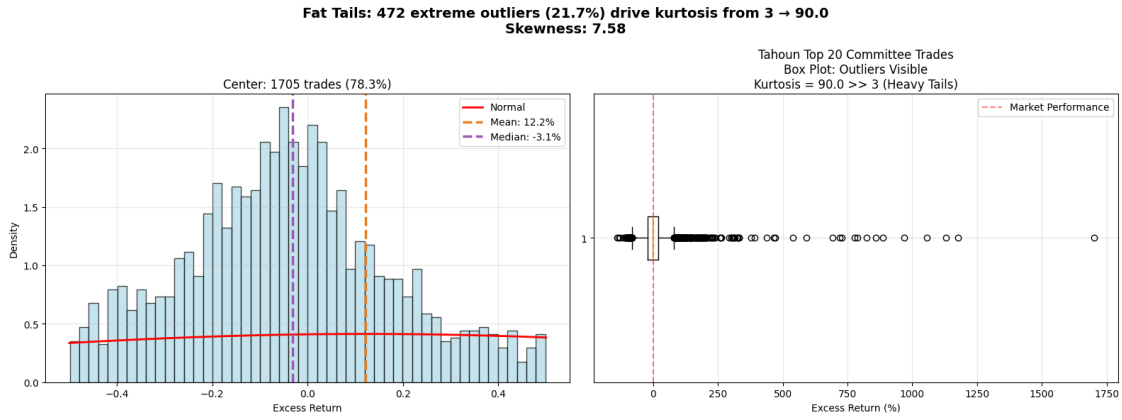


Figure 13: Skew and kurtosis of Tahoun top 20 committee excess returns, histogram centered about ± 50 percentage points and box plot of outliers.

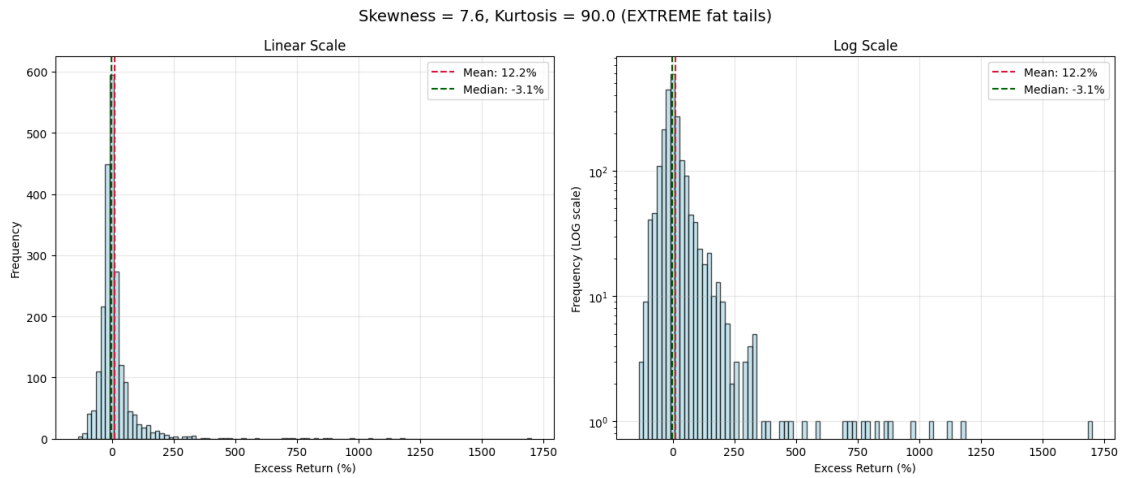


Figure 14: Linear and log scale histograms of Tahoun top 20 committee excess returns.

Table 16: Top 20 Committees Identified by OpenSecrets – Summary Statistics and Significance

Metric	Value
Num Committees	20
Sample size (trades)	2,199
S&P 500 benchmark (%)	12.49
Mean excess return (%)	+11.88
Median excess return (%)	−3.25
Excess return volatility (%)	96.25
Wilcoxon test statistic	1,150,895.00
P-value	0.975
Significance	ns
Effect size (rank-biserial r)	0.0484
Interpretation	Small

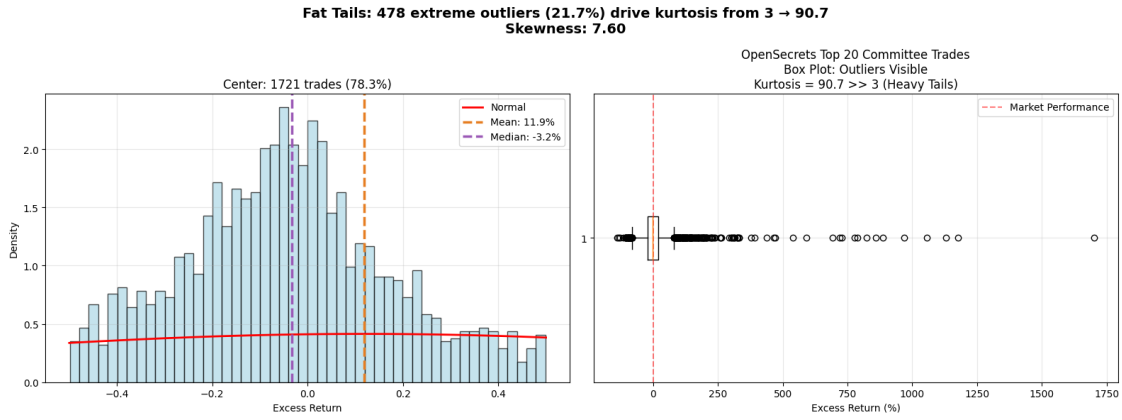


Figure 15: Skew and kurtosis of OpenSecrets top 20 committee excess returns, histogram centered about ± 50 percentage points and box plot of outliers.

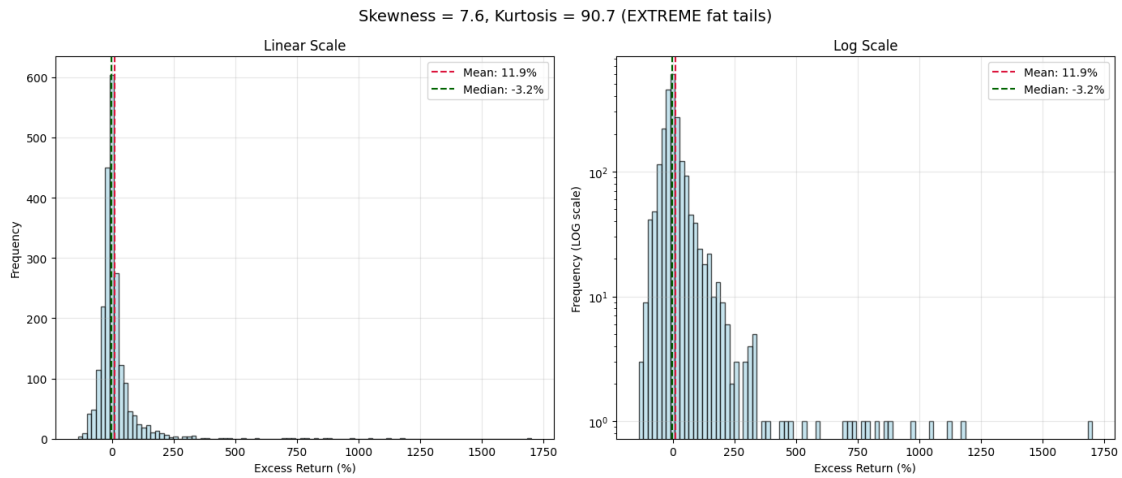


Figure 16: Linear and log scale histograms of OpenSecrets top 20 committee excess returns.

Tables 15 and 16 analyze trades from committees identified as particularly influential (listed below) by Tahoun 2013 and OpenSecrets, respectively. Both analyses reveal similar patterns. Tahoun's committees

averaged an excess return of 12.21%, a volatility of 96.64%, and were not significant at $\alpha = 0.05$ with a p-value of 0.956. OpenSecrets' committees averaged an excess return of 11.88%, a volatility of 96.25%, and were not significant with a p-value of 0.975. Figures 13–16 show that both datasets exhibit extreme kurtosis around 90, driven by roughly 21% of trades being outliers. The log scale histograms (Figures 14, 16) make the heavy-tailed distributions more apparent, and reveal considerable positive outliers despite the non-significant overall results. This suggests that while these influential committees show some exceptional trades, the combined committees performance is not superior to the market.

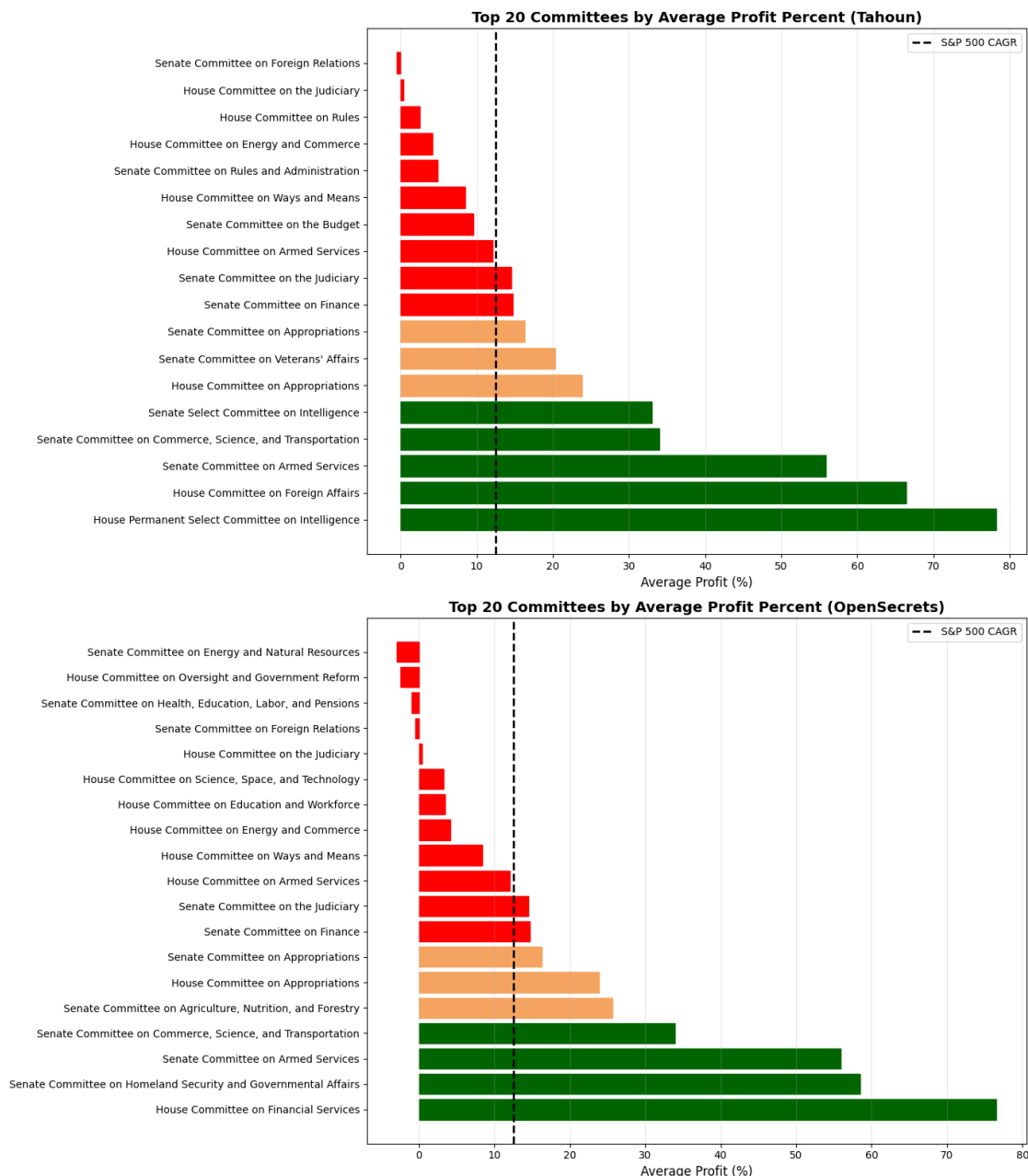


Figure 17: Bar chart for Tahoun and OpenSecrets top 20 committees by average profit.

Figure 17 shows that 8 of Tahoun's committees, and 10 from OpenSecrets fail to beat the market benchmark. While the best performers in Table 12 looked strong, we can observe that without cherry-picking those best performers, poorly performing committees balance out those who trade exceptionally well. This

reinforces the consistent pattern we have seen throughout the data, that some trades, Members, and committees do beat the market by a considerable amount, average and underperforming positions balance the outliers, resulting in overall insignificant returns.

4 Conclusion and Next Steps

This study assessed whether 12 years of data following the STOCK Act could provide insight into claims of Congressional Members exploiting informational advantages for personal gain. Building on related prior research as well as numerous articles and examples detailing evidence of suspicious trading activity, access to new and post-pandemic data has allowed for an updated analysis of the subject. The accusations, examples, and initial visualizations of top performers indicated possible wrongdoing, supported by statistical significance when comparing all congressional trades against S&P 500 market benchmarks. However, further analysis of annualized returns, Member specific portfolios, and examination of influential committee membership all provided insignificant results. Despite these overall findings, a noticeable and consistent trend appeared in almost every analysis. While many returns appear consistent with fair trading practices being near, or considerably worse than the market benchmark, a group of outliers saw substantial abnormal gains. This was apparent in the insignificant annualized returns, Member portfolios, distributions of returns, and committee performances. This smaller group of Members, committees, and trades were typically short term, high-returns that when set aside would beat the market by such large margins it would severely affect the rest of the data. These outliers are not enough to conclude wrongdoing, and more rigorous statistical testing would need to be implemented in order to investigate further. It does appear that most Members, whether or not they are given access to information to benefit upon, act in good faith, with some even seeing substantial losses. The outliers who consistently beat the market, and see gains far beyond those of top performing financial professionals, while also taking on considerably less risk, stand out due to the outstanding nature of their success. The question of additional legislation restricting congressional trading remains relevant, particularly given equity concerns when some Members achieve such exceptional returns.

If this research were to continue, it would be beneficial to tweak and expand on current tests, introduce more robust methods, non-static market metrics for certain significance tests, and test for significance in those groups of extreme outliers. Previous research also used a Carhart 4-factor model, which could be used to evaluate Member performance, but might be more difficult to implement. Further, exploration using machine learning to determine if stock performance can be predicted based on Member interest, or a collection of additional data on trades, Member portfolios, and active and past committee membership could strengthen the analysis and provide additional insight.

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